

EDUCATION/TRAINING REPORT

PRIVACY ACT STATEMENT

AUTHORITY: 10 U.S.C., Section 8013, Secretary of the Air Force; AFI 36-2406, Officer and Enlisted Eval Systems, and EOs 9397 and 13478 (SSN).

PURPOSE: To document effectiveness/duty performance history, promotion selection, reduction-in-force and other appropriate personnel actions.

ROUTINE USES: Disclosures generally permitted under 5 U.S.C. 552a(b) of the Privacy Act may specifically be disclosed outside the DoD as a routine use pursuant to 5 U.S.C., Section 552a(b)(3) and may be disclosed to other federal agencies. The "Blanket Routine Uses" apply.

DISCLOSURE: VOLUNTARY. However, nondisclosure may result in missing and/or inaccurate documents in the individuals master personnel records which may result in negative career impacting consequences.

SORN: F036 AF PC A, Effectiveness/Performance Reporting Records

I. IDENTIFICATION DATA (Read AFI 36-2406 carefully before filling in any item)

1. NAME (Last, First, Middle Initial) FIRTH, JORDAN A.	2. SSN 246-61-3834	3. RANK Maj	4. DUTY AFSC 92S0
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5. ORGANIZATION, COMMAND, AND LOCATION

Air Force Institute of Technology (AETC), Wright-Patterson AFB OH

6. PERIOD OF REPORT FROM: 17 Feb 2018 THRU: 24 Aug 2019	7. LENGTH OF COURSE 156 WEEK(S)	8. REASON FOR REPORT ☐ ANNUAL ☒ FINAL ☐ DIRECTED
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9. NAME AND LOCATION OF SCHOOL OR INSTITUTION

North Carolina State University (NCSU), Raleigh NC

10. NAME OR TITLE OF COURSE

Degree Program: PhD

II. REPORT DATA (Complete as applicable for final report)

1. AFSC/AERO RATING/DEGREE AWARDED Doctor of Philosophy	2. ☐ COURSE NOT COMPLETED (List reason in Item 4 below)
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3. DISTINGUISHED GRADUATE ☐ YES (List criteria in Item 4 below) ☒ NO DG PROGRAM

4. DG AWARD CRITERIA/COURSE NONCOMPLETION REASON

III. COMMENTS (Mandatory)

ACADEMIC/TRAINING ACCOMPLISHMENTS

- **DEGREE AWARDED:** Doctor of Philosophy in Aerospace Engineering
- **FOCUS AREA:** Deployable Spacecraft Structures
- **GPA:** 3.89 on a 4.0 scale; successfully completed all PhD requirements
- **DISSERTATION:** "Deployment Mechanisms for Rollable Spacecraft Booms"
 - Designed 2 separate mechanisms to consistently and reliably deploy booms--advanced field of space structures
 - Validated in-space performance via two manned flights--proved devices will enable lighter cheaper satellites
 - Research recognized by NASA as game changing development--awarded \$180K grant for development/flight test
 - Led 14-mbr team on multi-disciplinary test program culminating in safe, successful flights & data for 4 papers
 - "Advanced Dual-Pull Mechanism for Deployable Spacecraft Booms" published in Journal of Spacecraft & Rockets

PROFESSIONAL QUALITIES (Bearing, appearance, conduct, fitness)

Ratee is geographically separated from evaluator.

- **AFIT Liaison Officer** for 6 NCSU officers--mentored to degree completion & strengthened relationships with ROTC
- **Outstanding communicator;** selected as guest lecturer for AFROTC on space exploration and 2 engineering classes
- **Collaborated with CE dept** to validate bridge repair--will be used by NC Trans Dept to repair 226 damaged bridges
- **Long-term mentor** to 6 undergrad engineering students--comm & problem-solving skills will prepare them for jobs
- **Ran dept-wide equip training**--created documentation, ID'd shortfall, oversaw \$15K upgrade restoring key test ability

OTHER COMMENTS

- **Article titled "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms: Ground Test"** accepted for publication in the Journal of Spacecraft and Rockets
- **Presented paper titled "Minimal Unpowered Strain-Energy Deployment Mechanism for Rollable Spacecraft Booms"** at the American Institute of Aeronautics and Astronautics SciTech Forum

IV. EVALUATOR

NAME, RANK, BR OF SVC, ORGN, COMD, LOCATION THOMAS H. BROWN, JR., Colonel, USAF Air Force Institute of Technology (AETC) Wright-Patterson AFB OH	DUTY TITLE Director, Civilian Institution Programs	DATE 7 Jan 20
	SSN 2382	SIGNATURE 